

AWS advises: “Use short-term, least-privileged credentials and implement restrictive IAM policies. If customers have any questions about the security of their accounts, they are advised to contact AWS support.”

Rapid7 recommends: “Avoid long-term credentials, use temporary roles, implement least-privilege access, monitor suspicious activity, and proactively scan code repositories for secrets.”

This case highlights the dual-use nature of open source tools: TruffleHog, designed to protect organisations by detecting leaked secrets, is now being exploited to compromise them, underscoring the need for proactive secret management and restrictive IAM policies.

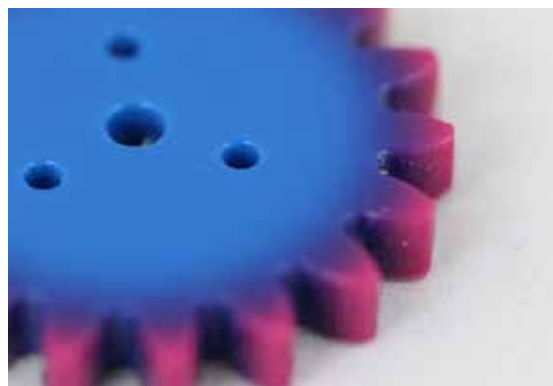
Python-based OpenVCAD will transform multi-material 3D printing

A new open source software tool, OpenVCAD, developed at the University of Colorado Boulder in the US, is transforming the way engineers create gradient, multi-material 3D-printed objects. The Python-based tool allows users to import the repository and start designing with a single line of code, democratising access to advanced 3D design without licensing barriers.

Charles Wade, a PhD student in computer science, said, “This is the first multi-material, code-based design tool that is widely available. Unlike traditional CAD software, where you’re forced to sketch everything out for each change and you

cannot represent graded materials, our tool allows users to change one small variable and watch the whole design update in an easy way.”

Developed in the Matter Assembly Computation Lab led by assistant professor Robert MacCurdy, OpenVCAD addresses the limitations of conventional CAD



tools, which typically define only an object’s outer boundaries and assume a single material inside. MacCurdy explained, “There’s certainly a history of multi-material design study and practice that existed well before OpenVCAD. But we believe the overhead of writing specific code for specific projects every single time prevents engineers from doing as much design as they could.”

OpenVCAD allows code-driven mapping of both shape and material distribution, supports seamless multi-material gradients, and has been demonstrated on printers capable of using up to five materials simultaneously. Applications span medical modelling, robotics, and structural engineering, enabling surgeons, roboticists, and engineers to design highly customised parts with unique mechanical properties.

Epaathsala releases generative AI-based ERP to enhance campus automation

Epaathsala, a *Times of India Group* company, has launched India’s first generative AI-driven open source ERP designed to transform campus automation. The platform combines generative and agentic AI to streamline institutional workflows—from admissions to alumni relations—while maintaining audit-ready compliance.

A standout feature is its conversational chatbot interface, which guides users through campus processes, simplifies data entry, and reduces errors. The system includes an AI teaching assistant that automatically generates course modules,

Samsung’s 7M-parameter AI model outperforms Gemini and DeepSeek

In a bold move that challenges the “bigger is better” philosophy of artificial intelligence, Samsung AI Labs in Montreal has unveiled the open source Tiny Recursive Model (TRM), a compact 7 million-parameter model that performs on par with systems thousands of times larger.

Released on GitHub alongside a research paper on arXiv, TRM represents a new direction in AI evolution, where intelligence is enhanced through recursive reasoning rather than computational scale. The model’s core innovation lies in its ability to refine its own answers iteratively, asking, “Is my answer good? If not, can I make it better?”

Despite its remarkably small size, nearly 10,000X smaller than leading LLMs such as OpenAI’s GPT, Google’s Gemini, and Anthropic’s Claude, TRM has demonstrated impressive benchmark results: 44.6% accuracy on ARC-AGI-1, 7.8% on ARC-AGI-2 (where most LLMs score below 5%), and 87% on Sudoku-Extreme. It even outperformed DeepSeek R1, Gemini 2.5 Pro, and o3-mini in reasoning tests, all while running efficiently on a laptop.

By simplifying the complex multi-network structures used in earlier recursive systems like the Hierarchical Reasoning Model (HRM), TRM proves that smarter, smaller architectures can achieve exceptional reasoning without relying on massive datasets or power-hungry infrastructure.

As Samsung emphasises, “Instead of building bigger models, one can build smarter, more efficient ones that think recursively.” The open source release of TRM signals a paradigm shift from closed, billion-parameter ecosystems to small, smart, and open AI models built for everyone.